
A Three-Layer Linear Ensemble with Round-Adaptive Calibration for Item Cold-Start Performance Prediction

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Abstract

We predict the probability that an AI subject answers an unseen benchmark item correctly using only the four text fields the platform provides. Our submission embeds each item with three large embedding models, reconstructs all label-dependent features at test time from shipped training statistics, and combines per-family predictors through a three-layer non-negative linear logit stack. A per-round calibrator refits benchmark-level intercepts on the revealed labels, and a temperature fitted by leave-one-benchmark-out (LOBO) validation corrects a systematic underconfidence of the stack. The full stack reaches 0.4252 validation BCE against 0.4380 for the best single member, and our ablations show that the layered combination, not any single model family, carries the gain. We also report two negative results we believe generalize, a member that helps under item-level validation but fails under benchmark-level shift, and an index-compression decision where the natural proxy metric badly overstated the downstream cost.

1 Introduction

The task is item cold-start prediction. Every test subject appears in the training matrix but no test item does, so the only usable signal is item content, subject identity, and the benchmark and condition labels. We treat this as soft binary cross-entropy regression from item text embeddings, and our design question is how to combine many partially redundant content-based predictors so the combination survives distribution shift to unseen benchmarks.

Our answer is deliberately simple. We build three independent prediction families, one per embedding model, and merge them with a three-layer *linear* logit stack fitted under non-negativity, plus a calibration layer that adapts each round to its revealed labels. A richer reference stack that adds an IRT member and a gradient-boosted inner stacker scores better on item-level validation (0.4218 vs. 0.4252), but we rejected those components. The hidden test is benchmark-shifted by construction, our benchmark-cold probes punished exactly the flexible components (Section 4), and a linear combiner has few enough parameters to leave little for shift to exploit.

Contributions, in the order a skimming reader should take away. (1) A three-layer non-negative linear ensemble over three embedding families that improves validation BCE from 0.4380 (best single member) to 0.4252, with each layer’s gain measured separately (Table 2). (2) A two-part calibration scheme, per-round benchmark intercepts fitted on the revealed labels plus a LOBO-fitted global temperature $\tau = 1.30$ that recovers $+0.007$ BCE on held-out benchmarks. (3) Negative results we believe generalize. An IRT member that helps under item-level validation ($+0.0011$) degrades severely on held-out benchmarks and was dropped; nearest-neighbor index recall is a poor proxy for the downstream cost of index compression; and bitwise train/serve consistency of embeddings is a real failure surface, not a formality.

Table 1: Feature manifest by leave-one-feature-group-out (LOO) group. Per family, one MLP variant is trained per withheld group and the layer-1 blend weights or prunes the variants. The first two groups are MLP input channels and the other eight form the 543-column dense block; rules separate the embedding channels, the label-free geometry groups, and the label-derived groups.

Group	Dims	Encodes	Why
item_embedding	4096	raw embedding of the templated item text	the primary content signal
subject_embedding	32	learned per-subject vector keyed by the hashed card	per-subject skill beyond an intercept
centroid_distance	256	squared distances to 256 fine k-means centroids	position in the embedding manifold
cluster_geometry	9	top-3 soft responsibilities, margin, entropy, typicality, log cluster size, coarse and fine ids	confidence of the cluster assignment
nn_geometry	3	mean neighbor similarity, top-to- k th gap, intrinsic-dimension estimate	local density and manifold shape
item_cluster	257	one-hot fine cluster plus its id	direct per-cluster effects
nn_label_derivatives	13	means and coverage of the subject’s neighbor pass rates at $k \in \{4, 8, 32, 64\}$, slopes, median, IQR, fraction passed	what this subject did on similar items
cluster_passrate	1	mean training pass rate of the item’s cluster	an item difficulty prior
cluster_subject	3	subject-vs-cluster gap, log observation count, responsibility-weighted subject cluster mean	subject skill relative to the region
counts_subject	1	count of the subject’s labels among the neighbors	reliability of the neighbor aggregates

2 Training data

We train on the public measurement-db corpus and curated no additional data. Prepared, it yields 4.50M (subject, item-variant) response rows covering 311,130 item variants, 906 subjects, and 16 benchmarks. An item variant is an (item, normalized condition) pair, matching the platform’s hidden snapshot. Labels are pass *rates* in $[0, 1]$, not binary outcomes (29% of rows are exactly 0 and 62% exactly 1), so all members are trained as cross-entropy regressors on soft targets and evaluated by mean soft BCE. All validation uses grouped 3-fold cross-validation with items as groups, so every reported out-of-fold (OOF) number is item cold-start by construction. Subjects and items are keyed by content hashes.

3 Method

Families and runtime features. Each family embeds the templated item text (benchmark, condition, item content) with one of Qwen3-Embedding-8B, llama-embed-nemotron-8b (with the bidirectional-attention protocol), or LGAI-Embedding-Preview. At test time a cold item arrives as raw text, so every feature must be recomputable from its embedding plus shipped training statistics. We ship, per family, k-means centroids (256 fine, 32 coarse) for cluster geometry features, a product-quantized (PQ-512) index over the 311k training-item embeddings for nearest-neighbor features, and a sparse subject-by-item pass-rate matrix for the label-derived aggregates of Table 1. The full dense block is 543 features. The shipped numpy-only runtime reproduces cluster assignments exactly and continuous geometry features to at most 1.4×10^{-2} (typically 10^{-4}); the index-dependent neighbor features are PQ-approximate, with their downstream cost quantified in Section 4.

Feature manifest. Table 1 lists the ten leave-one-feature-group-out (LOO) groups, which implement two pathways. The geometry groups locate the item in the embedding manifold without using labels, so they remain valid on an unseen benchmark. The label-derived groups inject collaborative signal, what this subject and the population did on similar items. The grouping gives the layer-1 blend one knob per pathway. The qwen blend zeroes the variants that withhold subject_embedding and item_cluster, while nemotron keeps all ten and lgai keeps nine.

Table 2: Validation soft BCE by ensemble layer (4.50M OOF rows, grouped 3-fold by item, lower is better). Each layer is evaluated on honest inner-CV inputs. The richer reference stack scores better under item-level validation but its extra components were rejected for benchmark-shift robustness; the previous-submission row is our earlier-generation architecture on the same data.

	qwen	nemotron	lgai
Best single MLP (all features)	0.4538	0.4446	0.4451
Forest only	0.4459	0.4380	0.4404
L1: LOO-MLP linear blend	0.4418	0.4295	0.4398
L2: + forest (family)	0.4387	0.4274	0.4367
L3: cross-family stack		0.4252	
Simple logit mean of families		0.4268	
Richer stack, +IRT +GBM inner stacker (rejected)		0.4218	
Previous submission (earlier architecture)		0.4365	

Subject metadata and semantic categories. The subject card is the platform’s `subject_content` field, a name line plus optional organization, parameter-count, release-date, and family lines. Training prep reproduces the platform’s rendering exactly, so the serve-time card hashes to the same key, which indexes the learned subject embedding and the subject’s pass-rate row. Card metadata thus enters the shipped system only as identity. Structured metadata tables for subjects and benchmarks (organization, parameters, release date, and family for 956 model variants, topic and age for the 16 benchmarks) fed our earlier-generation architecture and development analyses, not the shipped members. We also built a 15-category semantic taxonomy of item variants (skill, modality, and safety axes), an LLM-assisted but deterministic assigner over the runtime item fields. It served category-stratified validation during development and is not a runtime feature.

Members. Each family trains two member types. First, a bank of gated-GLU MLPs, one variant per withheld group of Table 1, which decorrelates the bank and lets the stack prune groups that do not earn weight. Second, a GPU random forest (depth 12, 128 trees) over the dense block plus a PCA of the embedding, ported to exact numpy inference (max deviation 9.6×10^{-8} from the training-time model). Members the selection stage rejected (XGBoost, factorization machines, kNN, denoising autoencoders, 1-D CNNs, logistic regression, IRT) are discussed in Section 4.

The three-layer linear stack. All combination is $z = \sum_j w_j \text{logit}(p_j) + b$ with $w_j \geq 0$, fitted by L-BFGS-B on soft BCE. Layer 1 blends each family’s LOO-MLP bank and prunes zero-weight members. Layer 2 blends the pruned MLP with the forest within each family. Layer 3 blends the three family probabilities (weights 0.64 nemotron, 0.25 lgai, 0.15 qwen). Each layer is fitted on honest inner-CV predictions of the layer below, so no layer fits on predictions that saw its rows.

Calibration and adaptive labeling. At test time two corrections sit on top of the stack. A global temperature τ sharpens the final logit, fitted offline by LOBO validation (Section 4). Then a per-round calibrator refits intercepts on the round’s revealed labels, with partial pooling. It fits a global intercept, an extra shift for benchmarks unseen in training, and per-benchmark intercepts shrunk toward their group value (ridge strengths 20/10/20), all by Newton steps on BCE. The offline temperature owns slope, which a handful of labels cannot estimate; the online intercepts own level, which they can. Our acquisition function requests labels with a dual-pool stratification that targets roughly 80% unseen-benchmark labels, because an extreme-value top- K selection cannot be tuned to an interior mix by score scaling alone.

4 Results and ablations

Ensemble depth carries the gain. Table 2 reports BCE at each layer. Within a family, blending the LOO-MLP bank beats the best single MLP by 0.005 to 0.015, and adding the forest gains another 0.002 to 0.003. Crossing families gains a further 0.0022 over the best family (nemotron, 0.4274) and 0.0016 over an unweighted logit mean. Every layer contributes, and the largest single step

is the within-family LOO blend, not the cross-family merge. The fitted family weights are far from uniform (0.64 nemotron), consistent with the three families converging on similar member architectures and differing mainly through their embeddings.

Member selection, and a warm/cold reversal. Under non-negative fitting on the full member pool, only the MLP bank and the forest receive weight; XGBoost, logistic regression, factorization machines, autoencoder and CNN members all go to zero as collinear with the survivors. A subject-embedding IRT member improved the stack by +0.0011 under item-level validation, but on benchmark-cold probes its BCE degraded by +0.09 to +0.42 depending on benchmark (three benchmarks probed), so we excluded it. A kNN member added only +0.0002, its neighbor signal already captured by the forest’s neighbor features. Item-level validation overrates members whose signal routes through within-benchmark structure.

Compressing the neighbor index, and a misleading proxy. The three $311k \times 4096$ indices cost 15 GB at full precision and 3.8 GB even at int8. We compared int8 (recall@64 of 0.991, 1.28 GB per family), PQ-512 (0.825, 164 MB), and PCA, which is disqualified outright (even 2048 dimensions keeping 98.6% of variance reaches recall 0.589; neighbor structure lives in low-variance directions). Recall suggested PQ-512 would be highly damaging. Downstream it is not. Replacing one family’s 17 index-dependent feature columns with PQ-derived values and re-scoring the fixed models moves that family’s forest by +0.0024 but the final stack by only +0.0009 BCE, because the stack’s weight concentrates on features the index does not touch. We shipped PQ-512.

LOBO temperature. For each benchmark we fit the BCE-optimal temperature on the stack logit and evaluate each benchmark with the median of the *other* benchmarks’ temperatures, so the evaluation is honest to an unseen benchmark. The shipped value is $\tau = 1.30$ with a held-out gain of +0.0066 BCE averaged per benchmark (+0.0074 row-weighted; 16 benchmarks, 4.50M rows; per-benchmark optima range 0.96 to 1.55). The direction surprised us. The stack is *underconfident* on held-out benchmarks, and a hypothesis consistent with the fitted weights is attenuation. Each layer is fitted on noisy inner-CV inputs, which biases slopes downward, and the layer-1 weight sums are indeed below one (0.90 to 0.97 across families). Member predictions on a held-out benchmark still come from models that saw its sibling items, so this τ is a floor on the true cold correction; we ship it as fitted.

5 Failure modes and patterns

Item-level validation is optimistic under benchmark shift. The predict-the-mean baseline scores 0.63 on our validation set, and our stack scores 0.4252, but probes that hold out entire benchmarks show member quality collapsing toward (and for IRT, beyond) the base rate. This single observation is why we prefer the linear combiner over the better-scoring richer stack, drop members that only help warm, refit intercepts every round, and temper the final logit. An out-of-distribution detector based on embedding distance was uninformative.

Train/serve embedding consistency is a real failure surface. Our features are only valid if the serve-time encoder reproduces the cached training embeddings. A minor-version library upgrade between training and serving changed the forward pass of one encoder family (identical weights and token ids, cosine to cache 0.49) and silently flipped a second family to bidirectional attention through a global attention-mask patch intended for one model. We caught both only by re-embedding training items at serve settings and comparing to the cache (the healthy families reproduce at cosine above 0.999). The fix scoped the mask patch to its model and re-embedded the broken family’s full cache under the serving stack, restoring consistency by construction. We recommend cache-reproduction checks as a standard preflight for any submission that ships embedding-derived features.

Limitations. All ablation numbers are item-grouped validation and overstate benchmark-cold performance; LOBO probes reuse members trained with within-benchmark siblings, so cold estimates are floors. The temperature and calibrator address level and slope but not feature-level transfer. We tuned the acquisition design in simulation against an assumed category structure, not measured platform behavior.